Hiba Altaf

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EDUCATION

Columbia University *New York, NY*

Bachelor of Science in Computer Science *Expected May 2027*

Relevant Coursework: Data Structures and Algorithms, Introduction to Databases, User Interface Design, Machine Learning, Artificial Intelligence, Advanced Programming in C, Fundamentals of Computer Systems, Probability

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, Go, TypeScript, SQL, C, C++

Frameworks & Tools: React, Node.js, PostgreSQL, Spring Boot, Kubernetes, AWS, Docker, LangGraph, Redis, CI/CD, Git

PROFESSIONAL EXPERIENCE

Oracle *June 2026 - September 2026*

Software Engineer Intern Redwood City, CA

- Engineered extensions to a **Python agent harness** and memory framework for an **autonomous** bug-resolution system, converting agent mistakes into reusable context and improving bug-resolution rate by 22%.
- Designed the framework's **relational data model** in Oracle DB and deployed the service to a **Kubernetes** beta environment on **OCI**; expanded integration and **CI/CD** test coverage by 25%.
- Collaborated with product managers and engineers to design and implement an automated customer-environment migration workflow, building its **agents**, **tools**, **policies**, and orchestration.

Amazon *May 2025 - August 2025*

Software Engineer Intern Seattle, WA

- Built and shipped a **full-stack** product comparison experience using **TypeScript** and **Java** across three Alexa device classes for an Alexa shopping platform serving 1M+ daily customers.
- Implemented a **Java Spring Boot** orchestration layer and **REST API** integrated with Alexa's core shopping flow, reducing response latency by 1.5–2 seconds and validating components and services through **unit and integration tests**.
- Partnered with a **UI** designer and product manager to design and prototype the product comparison interface in **Figma**, incorporating feedback to refine layouts and interaction patterns before implementation.

Amazon *May 2024 - August 2024*

Software Engineer Intern Seattle, WA

- Developed and evaluated an **LLM**-based product-title classification pipeline for Alexa, achieving 76% accuracy.
- Implemented the pipeline using **AWS EC2, S3, DynamoDB,** and **Docker**, processing 20,000+ JSON catalog records.
- Diagnosed and resolved a multi-item add-to-cart defect in a **Java** Spring Boot service supporting Alexa shopping.

Trillium Trading *July 2023 - May 2024*

Software Engineer Intern New York, NY

- Built a **TypeScript** and **React** training web application for 50+ day traders.
- Programmed an **NLP**-based search engine to retrieve data from 1,200+ company pages using Confluence **REST API.**

RESEARCH EXPERIENCE

Columbia University Electrical Engineering Department *August 2024 - November 2025*

Research Assistant *New York, NY*

- Developed a Python **agentic** code-analysis pipeline to detect and repair false sharing performance bugs in **multithreaded C/C++** codebases, integrating **RAG**, vector search, **KNN** retrieval, and iterative evaluation.
- Achieved 75% detection accuracy; generated fixes improved runtime in 56% of evaluated benchmarks.

LEADERSHIP EXPERIENCE

Columbia Space Initiative *October 2024 - May 2026*

Co-Lead *New York, NY*

- Led a 25-person engineering team developing a **mixed-reality** astronaut interface in **Unity**/**C#** for NASA's SUITS Design Challenge, integrating voice, hand, and gaze controls for navigation and field operations with Magic Leap 2.
- Created software components spanning **computer vision, speech interfaces**, and **SQLite**-based state management.

PROJECTS

Vaylo Fitness: Designed and **shipped** a **full-stack** fitness application with a Swift iOS client and **Python**/**LangGraph** backend. Built the client interface and state management; integrated **Supabase**/**PostgreSQL**, **Redis**, and Apple HealthKit to deliver personalized workout and nutrition recommendations.